CSC-343: Midterm Exam

(Uniformed Search Review Questions)

- **Show all work** for full credit. Answers without supporting steps or reasoning will receive little or no credit.
- Present your work in a **clear, logical, and organized manner**. Do not scatter steps with arrows or disconnected notes.
- **Clearly indicate your final answers** (e.g., box or highlight them).
- For proofs or problems, state all assumptions explicitly and justify each step.
- Only non-graphing calculators are permitted. No other electronic devices (including cell phones) are allowed.
- Use a pencil or erasable pen for your responses.
- For questions requiring responses that are not computational, you must use complete sentences.
- Any extraneous commentary about exam content will result in a reduced professionalism grade. If you have concerns, ask clarifying questions instead.
- Scratch paper is for your use only. It will be collected but not reviewed. Transfer all final work to this exam.
- Your responses should reflect the notation and techniques presented in class. While alternative conventions exist, consistency with course material is required for full credit.

By signing below, I certify that the work on this exam is my own without the use of external resources.

_____ _____

Printed Name Signature

Short Answer

1. What is the "frontier" in a search algorithm, and why is it important?

2. Explain why DFS is not complete in infinite state spaces, while BFS is complete in finite state spaces.

3. What is the key difference between Tree Search and Graph Search? When would you choose one over the other?

4. Explain the tradeoff that Iterative Deepening Search makes. Why is the redundant work at shallow levels acceptable?

5. What does it mean for a search algorithm to be "optimal"? Give an example of an optimal algorithm and a non-optimal algorithm from the uninformed search strategies.

6. Why might DFS be preferred over BFS in some situations, despite not guaranteeing optimal solutions?

True or **False**

1. Breadth-First Search is guaranteed to find the optimal solution when all edge costs are equal.

2. Depth-First Search requires less memory than Breadth-First Search because it only needs to store the current path and unexplored siblings along that path.

3. In graph search, a node is added to the explored set when it is first generated (added to the frontier).

4. Iterative Deepening Search has the same asymptotic time complexity as Breadth-First Search.

5. Tree search can visit the same state multiple times, while graph search visits each state at most once.

6. Uniform-Cost Search expands nodes in order of increasing depth from the start state.

7. In a search tree with branching factor b, solution depth d, and maximum depth m, BFS has space complexity $O(b^d)$ while DFS has space complexity $O(bd)$.

Other Questions

Question **1** and **2** refer to the following state space graph where `S` is the start state and `G` is the goal state. Edges are unlabeled (unit cost). As is typical from class, children are considered left-to-right.

```
        S
       /|\
      / | \
     A  B  C
    /|  |  |\
   D E  F  G H
    |     |
    G     G
```

1. Execute BFS. Show the contents of the frontier and explored set after each iteration until the goal is found. Use the format:

Iteration	Node Expanded	Frontier (front → back)	Explored Set
1		S	∅

What is the solution path found? Is this the optimal solution in terms of path length?

2. Execute DFS. Show the contents of the frontier and explored set after each iteration. Clearly indicate the top-of-stack using the rightmost element. What solution path does DFS find? Compared to the BFS solution, which found a shorter path?

3. Consider the following undirected, weighted graph with start `S` and goal `I`:

```
          S
         /|\
       2/ |3 \1
       /  |   \
      A   B    C
     /|   |    |\
   4/ |2  |5   |1 \6
   /  |   |    |   \
  D   E   F    G    H
       \       |
        \3     |2
         \     |
          \   /
            I
```

Execute Uniform-Cost Search, showing the priority queue contents (with path costs) after each node expansion. Break ties alphabetically.

Iteration	Node Expanded	Path Cost	Priority Queue
1		0	S

What is the optimal path and its total cost?

4. Consider the following tree:

```
        S
      / | \
     A  B  C
    /|  |  |\
   D E  F  G H
     |     |
     I     J (GOAL)
```

Execute Iterative Deepening Search, showing which nodes are visited at each depth limit (0, 1, 2, 3). As is typical from class, children are considered left-to-right.

Depth Limit	Nodes Visited (in order)	Goal Found?
0	S	No.

If this tree uniformly had branching factor $b = 3$ and the goal was at depth $d = 3$, calculate the total nodes visited by IDS.

5. Consider a search problem with branching factor $b = 4$, solution depth $d = 5$, and maximum depth $m = 10$.

- Calculate the worst-case number of nodes generated by BFS to find a solution at depth d. Express as both a formula and a numerical value.
- Calculate the worst-case number of nodes generated by DFS if it explores to maximum depth m before finding the solution.